

Project OZONE

Senior Flight Computer – Technical Documentation

UNSW Rocketry
30 km Sounding Rocket Avionics

*Named for the ozone layer, whose peak concentration occurs at 30–32 km altitude,
the maximum rated altitude of this board.*

Document revision: 1.0
Board revision: 1.0
Date: April 2026
EDA tool: KiCad 10
Fabrication: JLCPCB
Board designer: Dash

Repository: https://github.com/DashiellRussell/UNSW_Rocketry_Senior_Flight_Computer



Contents

1	Executive Summary	3
2	Board Overview	4
3	Introduction	8
3.1	Design Constraints	8
4	System Architecture	9
4.1	Block Diagram Overview	9
4.2	MCU: STM32L452RET6	9
4.2.1	Clock Configuration	9
4.2.2	Decoupling Strategy	9
4.2.3	Reset and Boot	9
5	Power System	11
5.1	Power Chain	11
5.2	Reverse Polarity Protection	11
5.3	Overcurrent Protection	11
5.4	Buck-Boost Regulator: TPS63060	11
5.5	LDO: SPX3819M5-L-3-3	12
5.6	Power Status LEDs	12
5.7	Battery Voltage Sensing	12
6	Sensor Suite	13
6.1	Barometers: MS5611 x2	13
6.2	High-g Accelerometer: H3LIS331DL	13
6.3	Low-g Accelerometer: LIS3DH	14
6.4	SPI Bus Considerations	14
7	Pyrotechnic System	15
7.1	Architecture Overview	15
7.2	Power Gating: Arm Circuit	15
7.3	Per-Channel Firing Circuit	15
7.4	Ground Isolation: PYRO_GND	15
7.5	Safety Architecture	16
8	Communications	17
8.1	USB-C	17
8.2	UART: Bluetooth Header	17
8.3	SWD Debug Header	17
9	Data Logging	18
9.1	SD Card Interface	18
9.2	Logged Data Columns	18
10	Indication and Status	19
10.1	RGB Status LED	19
10.2	Standalone LEDs	19
10.3	Piezo Buzzer	19
11	Board Capabilities	20

12 Board Limits and Restrictions	21
13 Flight Operations: Electrical Procedures	22
13.1 Pre-Flight Checklist	22
13.2 Arm Sequence	22
13.3 Flight Sequence	23
13.4 Post-Flight Recovery	23
13.5 Ground Test Firing	23
14 Operating Procedures	24
14.1 First Power-On (New Board)	24
14.2 Programming via SWD	24
14.3 Connecting a Bluetooth Module	25
14.4 Replacing the SD Card	25
14.5 Battery Replacement and Charging	25
14.6 Bench Testing Sensors	25
14.7 Continuity Testing (Without Pyro Battery)	26
14.8 Board Storage and Handling	26
15 Firmware Programming Tips	27
15.1 Development Environment	27
15.2 Peripheral Initialisation Order	27
15.3 SPI Sensor Communication	27
15.4 Apogee Detection: 2-of-3 Voting	28
15.5 Power Good Monitoring	28
15.6 Battery Voltage ADC	28
15.7 SD Card Write Strategy	28
16 Pin Reference	29
17 Design Decisions and Rationale	31
18 Test Points	32
18.1 Power Rails	32
18.2 Pyro System	32
19 BOM Summary	33
20 Connectors	34
21 Revision History	35

1 Executive Summary

Project OZONE is the senior flight computer for UNSW Rocketry's sounding rocket programme, rated for missions up to 30 km altitude. The board detects flight events, deploys two parachutes via pyrotechnic channels, logs all flight data to a MicroSD card, and provides visual and audible status indication throughout the mission.

The design centres on an STM32L452RET6 microcontroller (ARM Cortex-M4, 80 MHz, 512 KB flash) running the full avionics state machine. Altitude is measured by dual MS5611 barometers capable of resolving pressure down to 10 mbar (the 30 km floor), while a dual-accelerometer suite (H3LIS331DL for high-g launch/landing detection, LIS3DH for low-g apogee voting) provides redundant inertial data. Apogee is determined by a 2-of-3 voting algorithm across barometric, accelerometric, and timer-based sources before any pyrotechnic channel fires.

Power is supplied by a 1S or 2S LiPo battery through a buck-boost converter (TPS63060, 3.6 V) followed by an LDO (SPX3819, 3.3 V). The two-stage regulation ensures switching noise does not corrupt barometer readings at 10 cm altitude resolution. Reverse polarity protection, overcurrent protection, and four hardwired power status LEDs provide robust power chain diagnostics without firmware involvement.

The pyrotechnic system uses opto-isolated (PC817) gate drive with low-side N-FET switching (AO3400A), powered by a separate external battery gated through a P-FET arm circuit (AO3401A). TVS protection, star-bonded ground isolation, and continuity sensing are built into each channel. No arm or fire controls exist on the PCB itself; arming is via an external key switch, and ground test firing uses a 10-second delayed wireless command.

Communications include USB-C (firmware updates and log download), a UART header for an external Bluetooth module (wireless preflight checks), and a 5-pin SWD header for programming and debug. The board is fabricated as a 4-layer PCB (approximately 60 mm × 60 mm) via JLCPCB, using 0603 passives throughout for hand solderability with a stencil. Total component cost is approximately \$55 AUD.

2 Board Overview

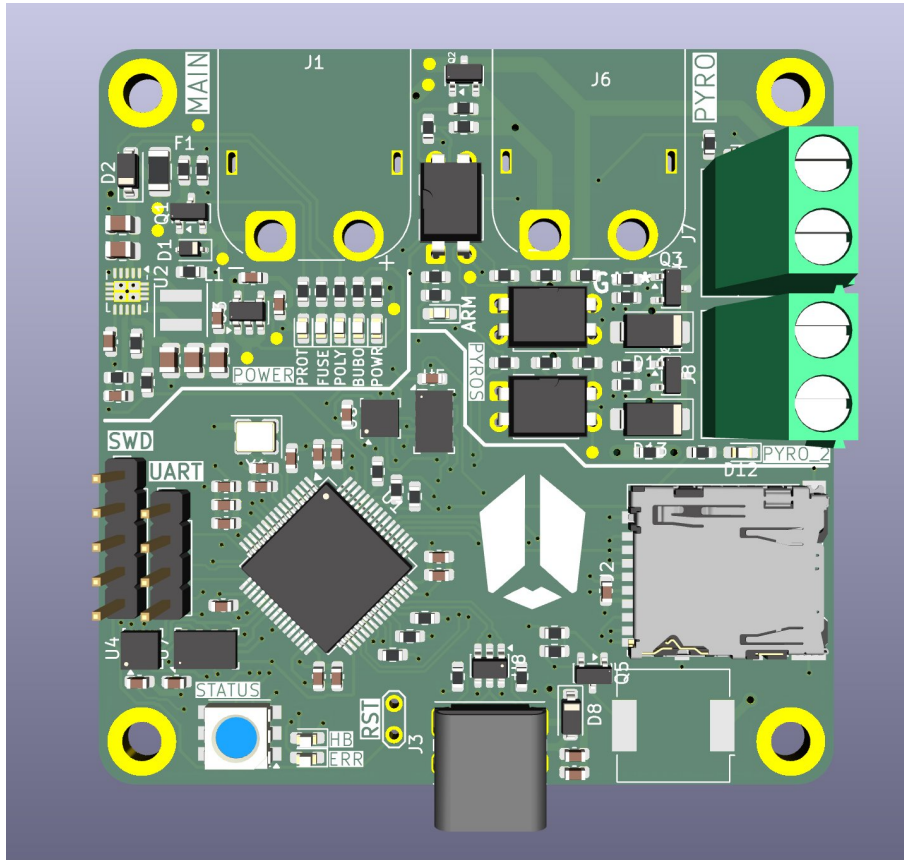


Figure 1: OZONE Senior Flight Computer, front side 3D render. Visible: STM32L452 MCU (centre), XT60 main battery connector (top left, J1), XT60 pyro battery connector (top right, J6), pyro screw terminals (right edge), SD card socket (right), USB-C (bottom centre, J3), SWD and UART headers (left), power status LEDs (upper left row), RGB status LED and heartbeat/error LEDs (bottom left).

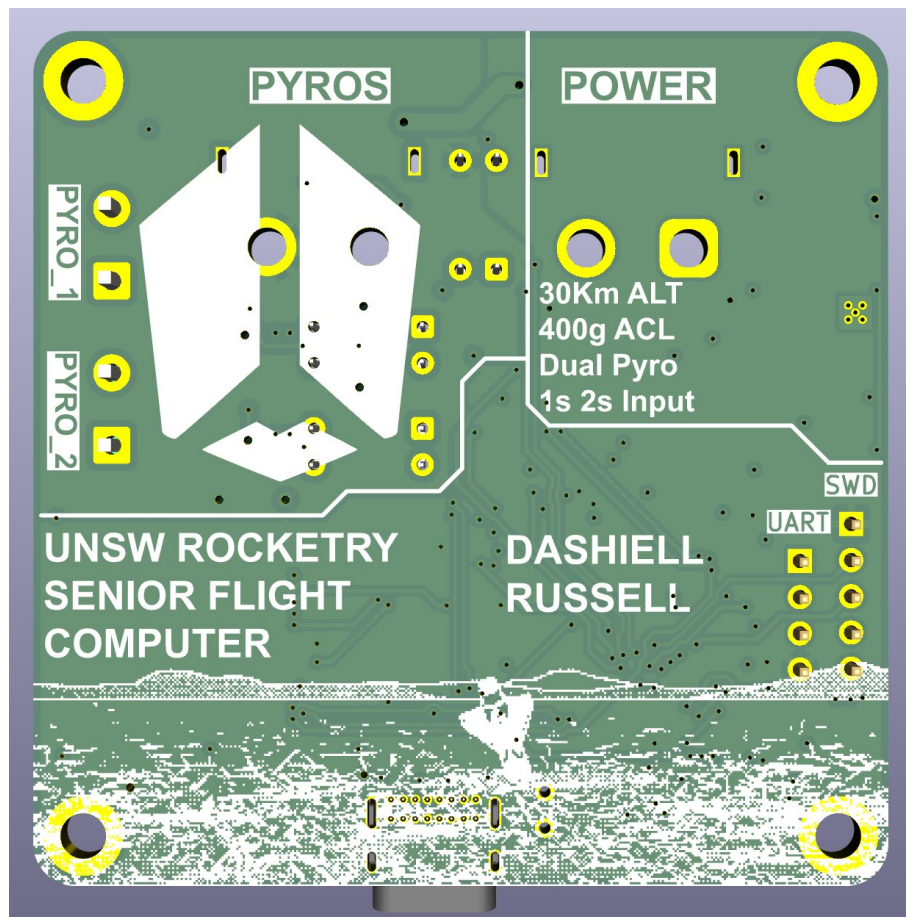


Figure 2: OZONE Senior Flight Computer, back side 3D render. Silkscreen includes board specifications (30 km altitude rating, 400 g acceleration rating, dual pyro channels, 1S/2S input compatibility), functional area labels, and designer credit.

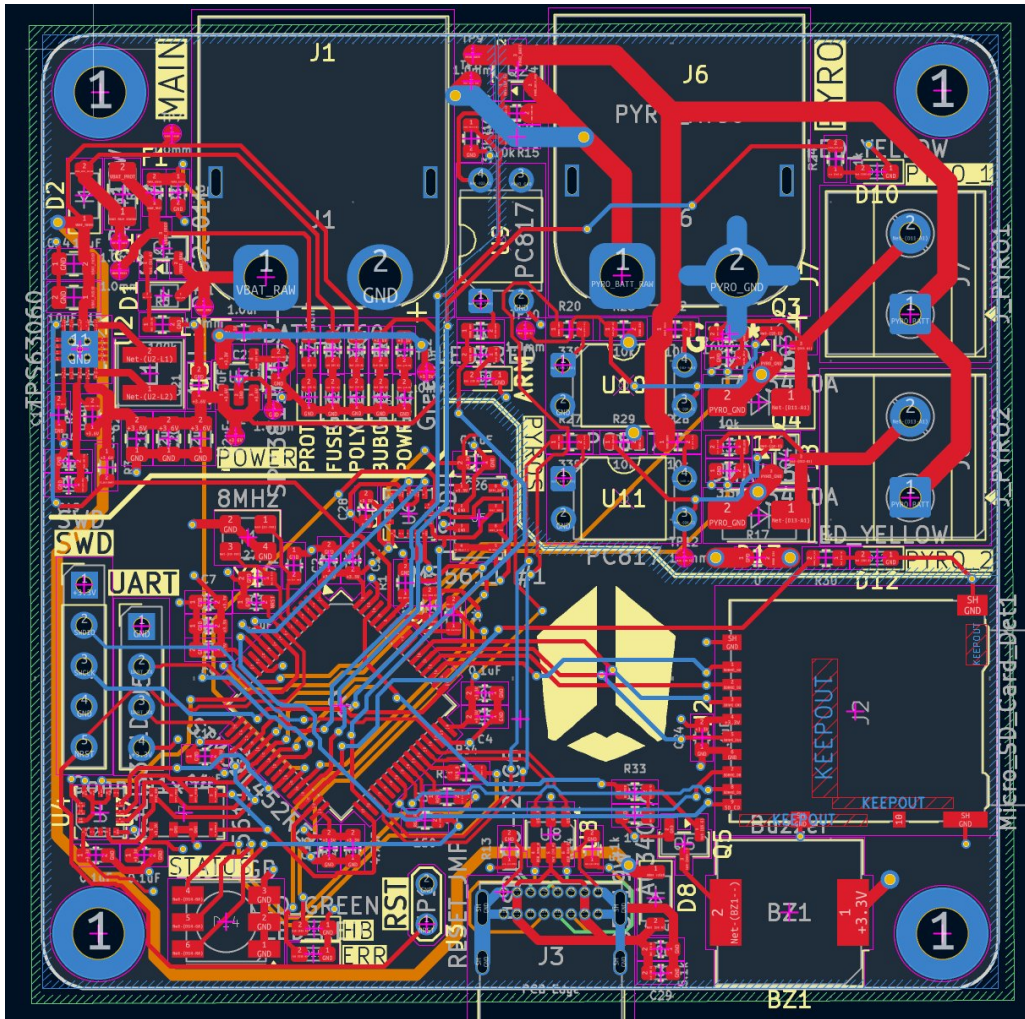


Figure 3: Front side PCB layout (all layers). Red traces are front copper (signal and power), blue traces are back copper, orange/yellow traces are inner layers. The power section occupies the upper left, pyro channels the upper right, MCU and sensors centre, and communications bottom left.

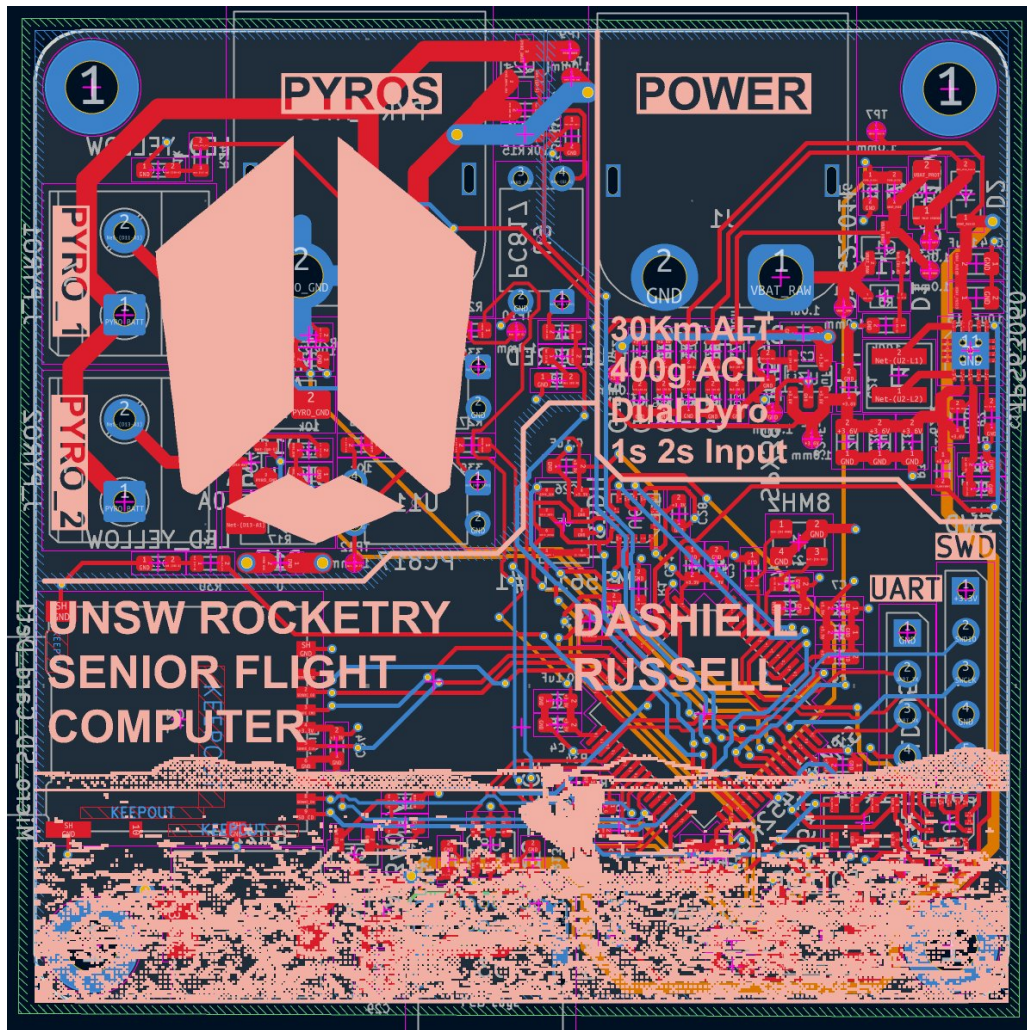


Figure 4: Back side PCB layout (all layers). The back side carries the UNSW Rocketry logo, board specifications, functional area labels (PYROS, POWER), and the designer name. Ground planes on inner layers provide shielding and low-impedance return paths.

3 Introduction

The OZONE Senior Flight Computer is a self-contained avionics board designed for sounding rockets up to 30 km altitude, developed by UNSW Rocketry. The project is named after the ozone layer, whose peak concentration occurs at 30–32 km, matching the maximum rated altitude of this board. The board is responsible for four mission-critical functions:

1. **Flight event detection:** launch, apogee, and landing via redundant sensor fusion.
2. **Parachute deployment:** two independent pyrotechnic channels (drogue and main).
3. **Flight data logging:** all sensor axes, pressure, temperature, pyro status, and flight events to a MicroSD card.
4. **Status indication:** visual and audible feedback for power state, arm state, continuity, errors, and post-landing recovery.

The board is designed around a single STM32L452RET6 microcontroller, with no development boards or modules integrated onto the PCB. All safety-critical controls (arming, firing) are deliberately kept off-board to prevent false triggering under the 200 g acceleration experienced during launch.

3.1 Design Constraints

Parameter	Value
Target budget (components)	\$50 AUD (actual: ~\$55 AUD)
Board dimensions	Approximately 60 mm × 60 mm
Layer count	4-layer PCB
Passive package	0603 throughout (hand-solderable with stencil)
Fabrication house	JLCPCB
Maximum acceleration	200 g (launch phase)
Target altitude	30 km

4 System Architecture

4.1 Block Diagram Overview

The system is organised into six functional blocks, each corresponding to a hierarchical schematic sheet:

Sheet	Contents
MCU (Root)	STM32L452RET6 with decoupling, crystal, reset, BOOT0, SWD header
Power	Battery input, reverse polarity protection, polyfuse, TPS63060 buck-boost, SPX3819 LDO, power status LEDs
Peripherals	SD card (SDMMC 4-bit), H3LIS331DL, LIS3DH, 2× MS5611 barometers on SPI1
Comms	USB-C with ESD protection, UART Bluetooth header, SWD debug header
Pyro	2-channel opto-isolated pyrotechnic system with arming, continuity sensing, and TVS protection
Indication	RGB LED, heartbeat/error LEDs, piezo buzzer

4.2 MCU: STM32L452RET6

Parameter	Value
Core	ARM Cortex-M4 with FPU, 80 MHz
Flash	512 KB
RAM	160 KB
Package	LQFP-64 (10 mm × 10 mm)
Supply range	1.71–3.6 V
Peripherals used	SPI1, SDMMC, USB FS, USART2, ADC1, TIM1, TIM17, GPIO

4.2.1 Clock Configuration

An external 8 MHz crystal (ECS-80-18-33-JGN-TR, 18 pF load capacitance) provides the HSE clock source. The PLL multiplies this to 80 MHz for the system clock. Load capacitors are 2× 27 pF, yielding an effective load of approximately 13.5 pF plus 3–5 pF of stray capacitance, matching the crystal specification.

4.2.2 Decoupling Strategy

Rail	Capacitors	Notes
VDD (pins 19, 32, 64)	3× 100 nF + 1× 4.7 µF bulk	One 100 nF per VDD pin
VDDA (pin 13)	1 µF + 100 nF + 10 nF	LC filter for clean analogue supply
VDDUSB (pin 48)	1 µF + 100 nF	USB peripheral power
VBAT (pin 1)	100 nF	RTC backup supply

4.2.3 Reset and Boot

- **NRST (pin 7):** 100 nF capacitor to GND for hardware debounce. A solder-bridge pad (replacing a tactile switch) allows bench reset when bridged. The internal pull-up holds NRST high during normal operation.

- **BOOT0 (PH3, pin 60):** 10 k Ω pull-down to GND. No jumper header. SWD provides all programming and debug capability, making DFU bootloader access unnecessary.

Note

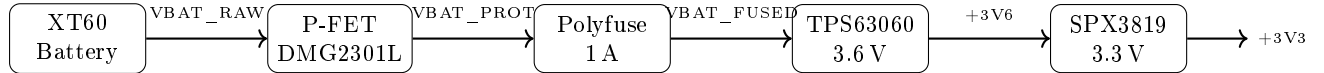
The tactile reset switch was replaced with solder-bridge pads to eliminate any risk of false reset under 200 g launch acceleration.

5 Power System

The power system accepts both 1S (3.0–4.2 V) and 2S (6.0–8.4 V) LiPo batteries through a single XT60PW-M connector.

5.1 Power Chain

The power path flows through five stages, each with a hardwired green status LED for visual debugging:



5.2 Reverse Polarity Protection

A P-channel MOSFET (DMG2301L-7, SOT-23) blocks current if the battery is connected backwards. A 5.1 V Zener diode clamps the gate-source voltage to -5.1 V, which is safely within the FET's ± 8 V V_{GS} rating while still fully turning on the device ($V_{GS(th)} = -0.4$ to -1.2 V).

Warning

The original design used an 8.2 V Zener, which would have allowed V_{GS} to reach -8.2 V at 2S, exceeding the absolute maximum rating. The 5.1 V Zener was chosen to provide adequate margin at all input voltages.

5.3 Overcurrent Protection

A 1 A PTC polyfuse protects the main power rail. This fuse is **not** placed inline with the pyrotechnic path. PTC fuses derate at elevated temperatures and would cause nuisance trips during flight.

5.4 Buck-Boost Regulator: TPS63060

Parameter	Value
Input voltage range	2.5–12 V
Output voltage	3.6 V (set by feedback divider: $R_{top} = 620$ k Ω , $R_{bot} = 100$ k Ω)
Switching frequency	2.4 MHz typical
Average current limit	2 A typical
Inductor	1 μ H, $I_{sat} > 3$ A, 4 mm $\times$ 4 mm
Input capacitors	2 $\times$ 10 μ F ceramic
Output capacitors	3 $\times$ 22 μ F ceramic
Power save mode	Enabled (PS/SYNC tied to GND)
Power good output	Open-drain, 100 k Ω pull-up to 3V6, routed to PB8 (PG_BUCKBOOST)

The output voltage is set by:

$$V_{OUT} = V_{FB} \times \left(1 + \frac{R_{top}}{R_{bot}}\right) = 0.5 \times \left(1 + \frac{620}{100}\right) = 3.6 \text{ V}$$

A 10 pF capacitor in parallel with the bottom feedback resistor provides high-frequency noise filtering on the FB node.

Power save mode (PS/SYNC = GND) improves efficiency at light loads while the board is idle on the pad. The converter automatically switches to full PWM mode when load current exceeds approximately 100 mA.

5.5 LDO: SPX3819M5-L-3-3

The SPX3819 generates the final 3.3 V rail from the 3.6 V buck-boost output. The two-stage approach (buck-boost then LDO) ensures switching noise from the TPS63060 does not corrupt the MS5611 barometer readings, which resolve altitude to 10 cm.

Parameter	Value
Input	3.6 V (from TPS63060)
Output	3.3 V
Dropout	340 mV typical at 500 mA
Bypass capacitor	10 nF on BP pin (noise reduction)
Input capacitor	1 μ F ceramic
Output capacitor	1 μ F ceramic

5.6 Power Status LEDs

Four hardwired (no GPIO required) green LEDs provide visual indication of power chain health:

LED	Rail	Resistor	Current	Indicates
PWR1	VBAT_PROT	4.7 k Ω	$\sim$ 1.3 mA at 8.4 V	Battery connected, correct polarity
PWR2	VBAT_FUSED	4.7 k Ω	$\sim$ 1.3 mA at 8.4 V	Polyfuse intact
PWR3	3V6	1 k Ω	$\sim$ 1.6 mA	Buck-boost regulating
PWR4	+3V3	1 k Ω	$\sim$ 1.3 mA	LDO output healthy

PWR1 and PWR2 use 4.7 k Ω resistors because the battery voltage varies from 3.0 to 8.4 V. Higher resistance keeps current safe at the upper end.

5.7 Battery Voltage Sensing

The main battery voltage is measured via a resistor divider connected to PC4 (ADC1_IN13). This allows firmware to monitor battery health throughout the flight and trigger low-battery warnings via the RGB LED.

6 Sensor Suite

All four sensors share a single SPI1 bus (PA5 = SCK, PA6 = MISO, PA7 = MOSI) with individual chip-select lines. Each sensor has 100 nF decoupling on its VDD pin.

6.1 Barometers: MS5611 x2

Parameter	Value
Part number	MS561101BA03-50
Quantity	2 (for redundancy and averaging)
Interface	SPI on SPI1
Chip selects	PB2 (CS_MS5611_1), PB12 (CS_MS5611_2)
Pressure range	10–1200 mbar
Resolution	24-bit ADC, 10 cm altitude resolution
Temperature	Built-in temperature output (satisfies mission temperature requirement)
Protocol select	PS pin tied to GND (SPI mode)

Note

The MS5611 was selected because it is the only affordable barometric sensor with a pressure floor low enough for 30 km altitude (~ 1.2 kPa). Common alternatives (BMP388, DPS368, SPL06) have 30 kPa minimum pressure limits, making them unsuitable.

6.2 High-g Accelerometer: H3LIS331DL

Parameter	Value
Part number	H3LIS331DLTR
Interface	SPI on SPI1
Chip select	PB0 (CS_H3LIS)
Interrupt	PC0 (INT1_H3LIS)
Range	± 100 g / ± 200 g / ± 400 g (selectable)
Package	TFLGA-16 (3 mm $\times$ 3 mm)
SPI clock max	10 MHz
Role	Launch detection, landing shock detection

Reserved pins: pin 10 and pin 16 are connected to GND; pin 15 is connected to VDD, per the datasheet.

6.3 Low-g Accelerometer: LIS3DH

Parameter	Value
Part number	LIS3DHTR
Interface	SPI on SPI1
Chip select	PB1 (CS_LIS3DH)
Interrupts	PC1 (INT1_LIS3DH), PC2 (INT2_LIS3DH)
Range	$\pm 2\text{ g}$ / $\pm 4\text{ g}$ / $\pm 8\text{ g}$ / $\pm 16\text{ g}$ (selectable)
Resolution	4 mg/LSB at $\pm 16\text{ g}$
Package	LGA-16 (3 mm $\times$ 3 mm)
Role	Apogee detection voter (gravity vector), low-g flight dynamics

The dual-accelerometer strategy provides complementary coverage. The H3LIS331DL handles the extreme g-loads of launch and landing but lacks the resolution to detect the subtle gravity changes at apogee. The LIS3DH (4 mg/LSB) provides a strong independent apogee voter.

6.4 SPI Bus Considerations

All four sensor chip-select lines default to high-impedance (floating) during MCU reset, as STM32 GPIOs power up in analogue mode. The sensors' internal pull-ups and the brief duration of this window ($\sim$ milliseconds) make this acceptable in practice, since the MCU does not reset during flight. However, firmware should explicitly drive all four CS lines high as an early initialisation step before configuring SPI1.

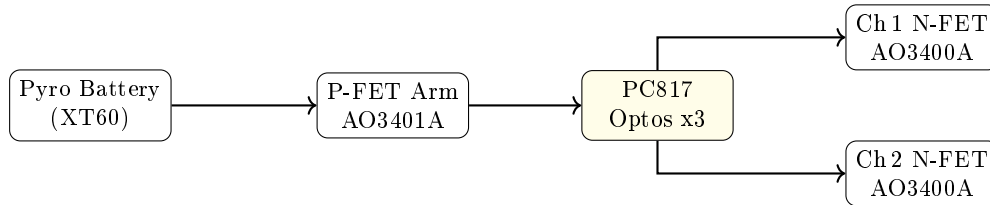
Table 1: SPI1 Bus Summary

Device	CS Pin	Max SPI Clock	Notes
MS5611 #1	PB2	20 MHz	Pressure + temperature
MS5611 #2	PB12	20 MHz	Redundant pressure + temperature
H3LIS331DL	PB0	10 MHz	Set multi-read bit (MS) for burst reads
LIS3DH	PB1	10 MHz	Set multi-read bit for burst reads

7 Pyrotechnic System

The pyro system is the most safety-critical subsystem on the board. It fires two independent e-match channels for drogue and main parachute deployment, powered by a separate external battery.

7.1 Architecture Overview



7.2 Power Gating: Arm Circuit

Component	Details
P-FET	AO3401A (SOT-23), gates the entire PYRO_BATT rail
Arm control	MCU GPIO drives a PC817 optocoupler; the opto output pulls the P-FET gate low to enable the pyro rail
Arm LED	Red LED illuminates when the pyro system is armed
Battery sense	Resistor divider from PYRO_BATT to MCU ADC (PC5 / ADC1) for pyro battery voltage monitoring

7.3 Per-Channel Firing Circuit

Each pyro channel consists of:

1. **Optocoupler (PC817):** Provides galvanic isolation between the MCU logic domain and the pyro power domain. The MCU drives the LED side; the phototransistor side controls the N-FET gate.
2. **N-FET (AO3400A, SOT-23):** Low-side switch. The e-match current flows from PYRO_BATT through the e-match, through the N-FET drain-to-source, and returns via PYRO_GND.
3. **Gate pull-down (10 kΩ):** Ensures the N-FET is **off** at power-up and during any transient condition.
4. **TVS diode (SMAJ15A):** Clamps inductive voltage spikes when the e-match fires.
5. **Continuity sensing:** A 10 kΩ / 3.3 kΩ resistor divider from the e-match terminal to ADC. The divider ratio ($\frac{3.3}{10+3.3} = 0.248$) scales the pyro battery voltage into the MCU's 0–3.3 V ADC range.
6. **Continuity LED (yellow):** Firmware-driven indication that the e-match is connected and healthy.

7.4 Ground Isolation: PYRO_GND

PYRO_GND is a separate net from the main board GND. The two ground domains are connected at a single point via a 0 Ω resistor (R10) in a star-bond configuration. This prevents high-current pyro return paths from injecting noise into the sensitive analogue and digital ground planes.

Warning

The continuity sensing divider bottom legs connect to **MCU GND**, not PYRO_GND. This is intentional. The ADC reference is relative to MCU GND, so the sense voltage must also be referenced to MCU GND for accurate readings.

7.5 Safety Architecture

Layer	Implementation
Hardware: isolation	Optocouplers provide galvanic isolation; N-FET gate pull-downs ensure channels are off at boot
Hardware: no buttons	No arm/fire buttons on PCB (false-trigger risk at 200 g)
Hardware: separate power	Pyro battery is independent from avionics battery
Hardware: TVS protection	SMAJ15A on each channel clamps flyback spikes
Software: 2-of-3 voting	Apogee must be confirmed by at least 2 of 3 independent sensor sources before drogue fires
Software: altitude lockout	No firing permitted below 200 altitude
Software: velocity lockout	No firing permitted below 3 m s^{-1} descent velocity
Software: ground test	Fire command requires 10-second delay for safety
External: arm switch	Physical key switch (off-board) must be closed to supply pyro battery

8 Communications

8.1 USB-C

Parameter	Value
Connector	USB-C receptacle (USB 2.0, 16-pin)
MCU pins	PA11 (USB_DM), PA12 (USB_DP)
CC pull-downs	$2 \times 5.1 \text{ k}\Omega$ on CC1 and CC2
ESD protection	USBLC6-2SC6 on data lines
VBUS protection	B5819W Schottky diode
Purpose	Firmware updates, configuration, log file download

8.2 UART: Bluetooth Header

A 4-pin header (VCC, TX, RX, GND) on USART2 (PA2 = TX, PA3 = RX) allows connection of an external Bluetooth serial module (e.g. HC-05 or RN4870). This enables wireless preflight health checks from a safe distance: sensor status, battery voltage, pyro continuity, and system state.

Note

No ESP32 or WiFi module is integrated. BLE serial via an external plug-in module provides sufficient bandwidth for preflight telemetry and avoids adding a second application processor.

8.3 SWD Debug Header

Pin	Signal	MCU Pin
1	+3V3	–
2	SWDIO	PA13 (pin 46)
3	SWCLK	PA14 (pin 49)
4	GND	–
5	NRST	Pin 7

Standard 5-pin, 2.54 mm pitch vertical pin header. Compatible with ST-Link V2/V3 and generic SWD adapters via dupont jumper wires.

9 Data Logging

9.1 SD Card Interface

Parameter	Value
Interface	SDMMC (4-bit bus, dedicated peripheral)
MCU pins	PC8 (D0), PC9 (D1), PC10 (D2), PC11 (D3), PC12 (CLK), PD2 (CMD)
Card detect	PC3 (SD_CD), active-low GPIO input
Connector	Micro SD card socket with card-detect switch
Decoupling	100 nF on VDD

SDMMC mode was chosen over SPI-mode SD access to avoid occupying the shared SPI1 bus. This required the STM32L452 (which includes the SDMMC peripheral) instead of the cheaper STM32L443.

9.2 Logged Data Columns

Column	Source	Rate
Timestamp	RTC / systick	Every sample
Altitude (m)	MS5611 #1 + #2 (averaged)	1 Hz
Pressure (mbar)	MS5611 #1, MS5611 #2	1 Hz
Temperature (°C)	MS5611 built-in	1 Hz
x/y/z acceleration (high-g)	H3LIS331DL	1 Hz
x/y/z acceleration (low-g)	LIS3DH	1 Hz
Pyro ch 1 status	Continuity ADC + fire state	1 Hz
Pyro ch 2 status	Continuity ADC + fire state	1 Hz
Flight event	State machine transition	On occurrence

Recommended write strategy: double-buffered DMA with a pre-allocated file. One DMA buffer fills from sensors while the other flushes to SD. This prevents missed samples during SD card write latency spikes. Pre-allocating the log file eliminates FAT table updates during flight.

10 Indication and Status

10.1 RGB Status LED

A common-cathode RGB LED (5050 PLCC-6 package) is driven by three PWM channels on TIM1:

Channel	MCU Pin	Timer	Resistor
Red	PA8	TIM1_CH1	150 Ω
Green	PA9	TIM1_CH2	100 Ω
Blue	PA10	TIM1_CH3	100 Ω

Suggested colour map (firmware-defined, changeable without hardware modification):

Colour	Meaning
Green steady	Idle, all sensors healthy
Blue breathing	Armed, waiting for launch
Red steady	Error / fault condition
Red blink	Low battery
Magenta	Flight mode active
Cyan	Landed, safe to approach
White flash	Pyro fired

10.2 Standalone LEDs

LED	Colour	GPIO	Purpose
Heartbeat	Green	PB6	Firmware running (blink pattern)
Error	Red	PB7	Sensor failure, SD error, or fault
Pyro 1 continuity	Yellow	(pyro sheet)	E-match 1 connected and healthy
Pyro 2 continuity	Yellow	(pyro sheet)	E-match 2 connected and healthy
Arm	Red	(pyro sheet)	Pyro system is armed

10.3 Piezo Buzzer

Parameter	Value
Part	CPT-9019S-SMT
Driver	AO3400A N-FET (low-side), gate driven by PB9 / TIM17_CH1
Gate resistor	10 k Ω
Purpose	Audible status indication and post-landing recovery locator

11 Board Capabilities

Capability	Details
Maximum altitude	30 km (limited by MS5611 pressure floor at ~ 1.2 kPa)
Acceleration range	± 400 g (H3LIS331DL max setting), ± 16 g (LIS3DH)
Battery compatibility	1S (3.0–4.2 V) and 2S (6.0–8.4 V) LiPo
Pyro channels	2 independent channels, each capable of 2 A at up to 8.4 V
Data logging	4-bit SDMMC, ~ 1 Hz minimum, event-driven flight events
Wireless telemetry	UART header for external BLE module
USB	Configuration, firmware update, log download via USB-C
Visual status	4 power LEDs + RGB LED + heartbeat + error + 2 continuity + arm LED
Audible status	Piezo buzzer (PWM-driven, recovery locator capable)
Debug	5-pin SWD header, multiple test points on power/digital/pyro nets

12 Board Limits and Restrictions

Limit	Detail
Input voltage	2.5–12 V absolute (TPS63060 input range). Practical range is 1S/2S LiPo (3.0–8.4 V)
Main rail current	1 A max (polyfuse trip point). Actual system draw is well below this
LDO output current	500 mA max (SPX3819 limit). System draw is ~50 mA
Pyro current	2 A per channel (N-FET and trace rating)
Altitude ceiling	~30 km (MS5611 lower pressure limit of 10 mbar)
Temperature	–40 to 85 °C (limited by the MS5611 operating range)
SD card size	Limited by FAT32 filesystem (32 GB max without exFAT support)
SPI bus speed	10 MHz max (limited by H3LIS331DL; MS5611 supports 20 MHz)
No CAN bus	CAN transceiver was removed from the final design; PB8/PB9 re-assigned
No on-board buttons	Arming and firing are exclusively off-board or via wireless command
Single SPI bus	All 4 sensors share SPI1, so sensor reads are sequential, not simultaneous

Warning

Never exceed 12 V on the main battery input. The TPS63060 absolute maximum is 12 V. A 3S LiPo (12.6 V fully charged) exceeds this limit and will damage the regulator. Use 1S or 2S only.

13 Flight Operations: Electrical Procedures

13.1 Pre-Flight Checklist

1. **Visual inspection:** Check for solder bridges, loose components, damaged connectors. Inspect pyro terminal screws.
2. **Power-on (avionics battery only):**
 - a. Connect 1S or 2S LiPo to the main XT60 connector.
 - b. Verify all four power status LEDs illuminate in sequence (PWR1 → PWR2 → PWR3 → PWR4).
 - c. Confirm heartbeat LED is blinking (firmware running).
 - d. Confirm RGB LED shows green steady (sensors healthy).
 - e. If any LED is off or the error LED is on, do **not** proceed. Debug via SWD or BT telemetry.
3. **Sensor verification (via BT telemetry):**
 - a. Connect to the Bluetooth module from a laptop/phone.
 - b. Request sensor health status. Verify both barometers, both accelerometers report valid data.
 - c. Verify altitude reading is approximately ground level.
 - d. Verify battery voltage is within expected range.
4. **SD card check:**
 - a. Ensure a formatted MicroSD card is inserted (card detect pin should read active).
 - b. Verify logging has started (firmware should confirm via BT or RGB colour change).
5. **Pyro system check:**
 - a. Connect e-matches to the pyro channel screw terminals.
 - b. Connect the pyro battery to the pyro XT60 connector (arm switch **open**).
 - c. Close the external arm key switch.
 - d. Verify the arm LED (red) illuminates.
 - e. Verify both pyro continuity LEDs (yellow) illuminate.
 - f. Request continuity ADC readings via BT. Both channels should show expected voltage.
 - g. Verify pyro battery voltage reading via BT.

13.2 Arm Sequence

1. All preflight checks passed.
2. External arm key switch is closed (supplies pyro battery to the board).
3. Firmware receives arm command via BT (or enters arm state based on flight state machine).
4. Firmware drives the arm optocoupler, turning on the P-FET, energising the PYRO_BATT rail.
5. Arm LED illuminates red. RGB LED transitions to blue breathing.
6. Board is now armed. All personnel retreat to safe distance.

13.3 Flight Sequence

1. **Launch detection:** H3LIS331DL detects sustained acceleration exceeding threshold. State machine transitions from ARMED to POWERED_FLIGHT.
2. **Powered flight:** All sensors logged at configured rate. RGB LED shows magenta (if visible).
3. **Apogee detection:** 2-of-3 voting from: barometer rate-of-change (both MS5611), LIS3DH gravity vector direction, and timer-based backup. When 2 sources agree apogee has occurred, drogue fires.
4. **Drogue deployment:** Firmware drives PYRO1_GATE high via optocoupler. N-FET turns on. E-match fires. Channel status logged. Altitude lockout (200) and velocity lockout (3 m s^{-1}) must be satisfied.
5. **Main deployment:** At a firmware-configured altitude (typically 500–1000 ft AGL), PYRO2_GATE fires the main parachute channel using the same mechanism.
6. **Landed:** Firmware detects sustained low acceleration and stable altitude. RGB LED transitions to cyan. Buzzer begins recovery tone.

13.4 Post-Flight Recovery

1. Follow the buzzer tone and/or GPS (if available externally) to locate the rocket.
2. **Before handling:** Open the external arm key switch to de-energise the pyro rail.
3. Verify the arm LED is off.
4. Disconnect the pyro battery.
5. Disconnect the main battery.
6. Remove the SD card for data download, or connect via USB to retrieve log files.

13.5 Ground Test Firing

For bench testing of pyro channels without a flight:

1. Connect e-match (or test load) to the desired channel.
2. Power the board and verify all checks pass.
3. Connect pyro battery and close arm switch.
4. Send fire command via BT telemetry.
5. Firmware imposes a mandatory **10-second delay** before firing, with countdown reported via BT.
6. After firing, verify channel status via ADC (continuity should read open/low after a successful fire).

Warning

Always follow your team's and range's pyrotechnic safety procedures. Never arm the pyro system unless e-matches are installed in a safe configuration. Never handle e-matches while the pyro system is armed. Treat all armed pyro systems as live.

14 Operating Procedures

This section provides step-by-step procedures for common board operations during development, integration, and field use.

14.1 First Power-On (New Board)

1. **Do not install the MCU yet** if hand-soldering. Verify power rails first.
2. Inspect the bare board under magnification for solder bridges, especially on fine-pitch components (TPS63060 WSON-10, SPX3819 SOT-23-5).
3. Connect a current-limited bench supply (set to 3.7 V, 200 mA limit) to the XT60 main battery connector.
4. Measure the following rails with a multimeter:
 - a. VBAT_PROT test point: should read approximately 3.7 V (confirms P-FET is on, correct polarity).
 - b. VBAT_FUSED test point: should read approximately 3.7 V (confirms polyfuse is intact).
 - c. +3V6 test point: should read 3.6 V $\pm 2\%$ (confirms TPS63060 is regulating).
 - d. +3V3 test point: should read 3.3 V $\pm 2\%$ (confirms SPX3819 LDO is regulating).
5. Verify all four power status LEDs are illuminated.
6. If any rail is incorrect, disconnect power immediately and inspect the failed stage.
7. Once rails are verified, install the MCU (if not already placed) and proceed to firmware bring-up.

14.2 Programming via SWD

1. Connect the ST-Link to the 5-pin SWD header using dupont jumper wires:
 - a. SWD pin 1 (+3V3) to ST-Link 3.3 V (or leave disconnected if board is self-powered).
 - b. SWD pin 2 (SWDIO) to ST-Link SWDIO.
 - c. SWD pin 3 (SWCLK) to ST-Link SWCLK.
 - d. SWD pin 4 (GND) to ST-Link GND.
 - e. SWD pin 5 (NRST) to ST-Link NRST (optional, but recommended).
2. Power the board via the main battery connector.
3. In STM32CubeIDE or OpenOCD, connect to the target. The MCU should be detected as STM32L452RE.
4. Flash firmware and verify using the built-in flash verification.
5. After flashing, the heartbeat LED (PB6) should begin blinking if the firmware includes a heartbeat task.

14.3 Connecting a Bluetooth Module

1. The UART header (J5) provides: pin 1 = VCC (3.3 V), pin 2 = TX (PA2), pin 3 = RX (PA3), pin 4 = GND.
2. Connect the Bluetooth module (e.g. HC-05 or RN4870) with crossed TX/RX lines:
 - a. Board TX (PA2) connects to module RX.
 - b. Board RX (PA3) connects to module TX.
3. Configure the Bluetooth module baud rate to match the firmware USART2 configuration (typically 115200 baud, 8N1).
4. Pair with the module from a laptop or phone. Use a serial terminal application to send and receive telemetry.

14.4 Replacing the SD Card

1. Power off the board by disconnecting the main battery.
2. Remove the existing MicroSD card from the socket.
3. Format the new card as FAT32 (for cards up to 32 GB). Cards larger than 32 GB require exFAT, which may not be supported by the firmware.
4. Insert the new card. The card detect pin (PC3) will register the insertion on next power-up.
5. Power on the board and verify that the firmware acknowledges the card (check via BT telemetry or RGB LED colour).

14.5 Battery Replacement and Charging

1. Always power off the board before disconnecting or connecting a battery.
2. The main battery (1S or 2S LiPo) connects via the XT60PW-M connector (J1). The connector is keyed and cannot be inserted backwards, but the P-FET protection circuit will block current if a non-keyed adapter is wired in reverse.
3. The pyro battery connects via the second XT60PW-M connector (J6). This battery should only be connected when the external arm key switch is open.
4. Always charge LiPo batteries using a dedicated LiPo charger. Never charge batteries while they are connected to the board.
5. Recommended minimum voltages before flight: main battery above 3.6 V (1S) or 7.2 V (2S); pyro battery above 3.6 V (1S) or 7.2 V (2S).

14.6 Bench Testing Sensors

1. Power the board and establish a BT telemetry link (or connect via SWD debugger).
2. Read the WHO_AM_I register from each sensor to verify SPI communication:
 - a. H3LIS331DL: WHO_AM_I (register 0x0F) should return 0x32.
 - b. LIS3DH: WHO_AM_I (register 0x0F) should return 0x33.
 - c. MS5611: Send a reset command (0x1E), then read PROM coefficients. Valid coefficients confirm communication.

3. Verify accelerometer readings at rest: the LIS3DH should show approximately 1g on the vertical axis and near-zero on the other two axes.
4. Verify barometer readings: both MS5611 sensors should report ambient pressure (approximately 1013 mbar at sea level) and a reasonable temperature.
5. If any sensor fails to respond, check the corresponding CS line, SPI bus connections, and decoupling capacitor.

14.7 Continuity Testing (Without Pyro Battery)

1. Connect a known-value test resistor (e.g. $2.2\ \Omega$ wirewound) across the pyro channel screw terminals in place of an e-match.
2. Connect the pyro battery and close the arm switch.
3. Read the continuity ADC value via BT telemetry. It should correspond to the expected voltage from the divider with the test resistor in circuit.
4. Repeat for the second channel.
5. Open the arm switch and disconnect the pyro battery when testing is complete.

14.8 Board Storage and Handling

1. Store the board in an antistatic bag when not in use.
2. Disconnect all batteries before storage.
3. Remove the SD card if the board will be stored for an extended period.
4. Avoid touching exposed pads and component leads. Handle the board by its edges or mounting holes.
5. The board uses 0603 passives throughout. Use appropriate ESD precautions (wrist strap, ESD-safe work surface) during handling and rework.

15 Firmware Programming Tips

15.1 Development Environment

- **Toolchain:** ARM GCC or STM32CubeIDE.
- **Programming interface:** SWD via ST-Link V2 or V3. Connect to the 5-pin SWD header with dupont wires.
- **HAL vs LL:** STM32 HAL is recommended for initial bring-up. LL (low-level) drivers can be used for time-critical paths (SPI sensor reads, timer interrupts).

15.2 Peripheral Initialisation Order

A recommended initialisation sequence to ensure safe power-up:

1. Configure system clock (HSE → PLL → 80 MHz).
2. **Immediately** configure all SPI CS pins (PB0, PB1, PB2, PB12) as GPIO outputs and drive them **high**. This deselects all sensors before SPI1 is configured.
3. Configure all pyro gate pins as GPIO outputs, drive **low**. Confirm arm pin is low.
4. Initialise SPI1 (mode 0 or mode 3 depending on sensor, clock ≤ 10 MHz for H3LIS331DL compatibility).
5. Initialise SDMMC and mount filesystem.
6. Initialise ADC1 for battery voltage sensing, pyro continuity sensing, and pyro battery sensing.
7. Initialise USART2 for Bluetooth telemetry.
8. Initialise TIM1 (RGB LED PWM) and TIM17 (buzzer PWM).
9. Read WHO_AM_I registers from all four sensors to verify SPI communication.
10. Begin heartbeat LED blink.
11. Enter idle state, report status via BT.

15.3 SPI Sensor Communication

- Both ST accelerometers use bit 0 of the first byte as the R/W flag (1 = read, 0 = write) and bit 1 as the multi-read (MS) flag.
- The MS5611 uses a command-based protocol: send a conversion command, wait for the ADC conversion time (~ 8.22 ms at $OSR=4096$), then read the result. Do **not** block during the conversion. Use a timer or state machine to poll.
- When reading from multiple sensors in sequence, deassert each CS before asserting the next. Allow at minimum the CS hold time specified in each datasheet (8 ns for H3LIS331DL).

15.4 Apogee Detection: 2-of-3 Voting

Implement three independent apogee voters:

1. **Barometric:** Rate of change of altitude (from averaged MS5611 readings) crosses zero and goes negative.
2. **Accelerometric:** LIS3DH z-axis transitions from negative (upward acceleration) through zero-g to positive (freefall).
3. **Timer backup:** If a configurable time has elapsed since launch detection and altitude is above a minimum threshold, vote for apogee.

Fire drogue only when ≥ 2 voters agree, **and** the altitude lockout (200) and velocity lockout (3 m s^{-1}) conditions are satisfied.

15.5 Power Good Monitoring

PG_BUCKBOOST (PB8) is an open-drain output from the TPS63060. Configure PB8 as a GPIO input. When the pin reads low, the buck-boost output has fallen out of regulation. Log this event and consider entering a safe state.

15.6 Battery Voltage ADC

The main battery voltage divider on PC4 (ADC1_IN13) scales the battery voltage into the 0–3.3 V ADC range. To convert ADC counts back to battery voltage:

$$V_{BAT} = \frac{ADC_{raw}}{4096} \times 3.3 \times \frac{R_{top} + R_{bot}}{R_{bot}}$$

where R_{top} and R_{bot} are the divider resistor values. Calibrate this against a known-good multimeter reading during bench testing.

15.7 SD Card Write Strategy

- Pre-allocate a large log file at startup to avoid FAT table updates during flight.
- Use double-buffered DMA: one buffer fills from sensors, the other flushes to SD.
- If a write fails, retry once; if it fails again, set the error LED and continue logging to the backup buffer (do not halt the flight state machine for an SD error).
- Flush and close the file on landing detection.

16 Pin Reference

MCU Pin	Port.Pin	Function	Net Label
1	VBAT	Power	VBAT
5	PH0	HSE_IN	Crystal
6	PH1	HSE_OUT	Crystal
7	NRST	Reset	NRST
8	PC0	GPIO input	INT1_H3LIS
9	PC1	GPIO input	INT1_LIS3DH
10	PC2	GPIO input	INT2_LIS3DH
11	PC3	GPIO input	SD_CD
13	VDDA	Power	+3V3 (analogue)
14	PA0	GPIO	INT1_H3LIS (alt)
16	PA2	USART2_TX (AF7)	UART_TX
17	PA3	USART2_RX (AF7)	UART_RX
20	PA4	GPIO output	–
21	PA5	SPI1_SCK (AF5)	SPI1_SCK
22	PA6	SPI1_MISO (AF5)	SPI1_MISO
23	PA7	SPI1_MOSI (AF5)	SPI1_MOSI
24	PC4	ADC1_IN13	VBAT_SENSE
25	PC5	ADC1_IN14	PYRO_BATT_SENSE
26	PB0	GPIO output	CS_H3LIS
27	PB1	GPIO output	CS_LIS3DH
28	PB2	GPIO output	CS_MS5611_1
29	PB10	GPIO output	PYRO1_GATE
30	PB11	GPIO output	PYRO2_GATE
33	PB12	GPIO output	CS_MS5611_2
34	PB13	GPIO output	PYRO1_CONT_LED
35	PB14	GPIO output	PYRO2_CONT_LED
36	PB15	GPIO output	PYRO_ARM
37	PC6	ADC / GPIO	PYRO1_CONT
38	PC7	ADC / GPIO	PYRO2_CONT
39	PC8	SDMMC1_D0 (AF12)	SDMMC_D0
40	PC9	SDMMC1_D1 (AF12)	SDMMC_D1
41	PA8	TIM1_CH1 (AF1)	RGB_R
42	PA9	TIM1_CH2 (AF1)	RGB_G
43	PA10	TIM1_CH3 (AF1)	RGB_B
44	PA11	USB_DM (AF10)	USB_N
45	PA12	USB_DP (AF10)	USB_P
46	PA13	SWD	SWDIO
49	PA14	SWD	SWCLK
51	PC10	SDMMC1_D2 (AF12)	SDMMC_D2
52	PC11	SDMMC1_D3 (AF12)	SDMMC_D3
53	PC12	SDMMC1_CK (AF12)	SDMMC_CLK
54	PD2	SDMMC1_CMD (AF12)	SDMMC_CMD
58	PB6	GPIO output	LED_HEARTBEAT
59	PB7	GPIO output	LED_ERROR
60	PH3	BOOT0	10k pull-down
61	PB8	GPIO input	PG_BUCKBOOST

62	PB9	TIM17_CH1 (AF14)	BUZZER
----	-----	------------------	--------

17 Design Decisions and Rationale

This section documents the key engineering trade-offs made during the design process for future reference.

1. **MS5611 over BMP388/DPS368/SPL06:** The MS5611 is the only affordable barometric sensor that reaches the 10 mbar pressure floor needed for 30 km altitude. All other candidates have 300 mbar (30 kPa) minimum operating pressure.
2. **Dual accelerometer (H3LIS331DL + LIS3DH):** The high-g sensor detects launch and landing but lacks resolution near apogee. The low-g sensor (4 mg/LSB) provides a strong independent apogee voter. Both use LGA-16 packages and the same ST driver ecosystem.
3. **H3LIS331DL over ADXL375:** The H3LIS331DL saved approximately \$5 AUD while offering the same LGA-16 footprint and compatible SPI interface.
4. **TPS63060 buck-boost + SPX3819 LDO (two-stage regulation):** A single buck-boost handles the wide 1S/2S input range. The downstream LDO removes switching ripple that would corrupt MS5611 pressure readings at 10 cm resolution.
5. **SDMMC over SPI-mode SD:** Avoids sharing the SPI1 bus with four sensors. Required upgrading from STM32L443 to STM32L452 (~\$5 AUD premium), but eliminates bus contention concerns.
6. **Opto-isolated pyro channels:** PC817 optocouplers provide galvanic isolation between the MCU domain and the high-current pyro domain, adding a hardware isolation layer beyond the separate ground domains.
7. **Separate pyro battery:** Keeps pyro firing current completely off the avionics supply. Simplifies power design and eliminates the risk of sensor supply droop during e-match ignition.
8. **No on-board buttons:** At 200 g, any tactile switch will false-trigger. Arming is via external key switch; ground test firing uses the 10-second delayed UART/BT command.
9. **No PTC fuse on pyro path:** PTC fuses derate at elevated temperatures, which would cause nuisance trips during flight. The pyro path relies on the external arm switch and N-FET gate control for protection.
10. **BOOT0 pull-down only (no jumper):** SWD provides complete programming and debug access. A BOOT0 jumper header adds unnecessary complexity and another potential failure point.
11. **PS/SYNC tied to GND:** Enables power-save mode on the TPS63060 for improved efficiency at light loads while idle on the launch pad. The converter automatically returns to full PWM mode when load increases during flight.
12. **External BT module over integrated ESP32:** A UART header with a plug-in BLE module avoids a second MCU, reduces board complexity, and allows the wireless module to be removed for flight if desired.

18 Test Points

The board includes test points for oscilloscope and multimeter access during development and debugging.

18.1 Power Rails

Test Point	Signal	Purpose
TP1	GND	Scope ground reference
TP2	VBAT_RAW	Battery input voltage
TP3	+3V6	Buck-boost output verification
TP4	GND	Second ground reference
TP5	+3V3	LDO output, measure ripple
TP6	+3V3	Additional 3.3 V access
TP7	GND	Additional ground reference

18.2 Pyro System

Test Point	Signal	Purpose
TP8	PYRO_GND	Pyro ground domain
TP9	PYRO_BATT	Pyro battery rail (after arm FET)
TP10	–	Pyro channel monitoring
TP12	–	Pyro channel monitoring

19 BOM Summary

Component	Part Number	Qty	AUD
MCU	STM32L452RET6TR	1	\$14.34
Barometer	MS561101BA03-50	2	\$13.86
High-g accelerometer	H3LIS331DLTR	1	\$11.16
Low-g accelerometer	LIS3DHTR	1	\$0.50
Buck-boost	TPS63060DSCR	1	\$2.27
LDO	SPX3819M5-L-3-3/TR	1	\$0.24
P-FET (polarity prot.)	DMG2301L-7	1	\$0.58
HSE crystal	ECS-80-18-33-JGN-TR	1	\$1.24
USB ESD protection	USBLC6-2SC6	1	–
Pyro optocouplers	PC817	3	–
Pyro N-FETs	AO3400A	2	–
Pyro arm P-FET	AO3401A	1	–
TVS diodes	SMAJ15A	2	–
Estimated total (incl. passives, connectors, LEDs)			~\$55

Note

The BOM exceeds the \$50 AUD target by approximately \$5. The primary cost drivers are the H3LIS331DL (\$11.16) and the dual MS5611 barometers (\$13.86 total). These are non-negotiable for the 30 km mission profile.

20 Connectors

Connector	Type	Purpose
J1	XT60PW-M	Main battery input
J2	Micro SD socket (with detect)	Data logging
J3	USB-C receptacle (16-pin, USB 2.0)	Firmware, config, log download
J4	5-pin header, 2.54 mm	SWD debug/programming
J5	4-pin header, 2.54 mm	UART Bluetooth module
J6	XT60PW-M	Pyro battery input
J7	Screw terminal / header	Pyro channel 1 (e-match)
J8	Screw terminal / header	Pyro channel 2 (e-match)
H1-H4	Mounting holes	Mechanical mounting

21 Revision History

Rev	Date	Changes
1.0	April 2026	Initial release as Project OZONE. Schematic complete, layout complete, ready for fabrication.

